Sampling and indexing

When to use

- You need to decide a sampling stride/FPS and ensure all downstream outputs (interval instructions, per-frame artifacts, etc.) cover the same frame range with consistent indices.

Core steps

- Read video metadata: frame count, fps, resolution.
- Choose a sampling strategy (e.g., every 10 frames or target ~10-15 fps) to produce sample_ids.
- Only produce instructions and masks for sample_ids; the max index must be < total_frames.
- Use a strict interval key format such as "{start}->{end}" (integers only). Decide (and document) whether end is inclusive or exclusive, and be consistent.

Pseudocode

```
import cv2
VIDEO_PATH = "<path/to/video>"
cap = cv2.VideoCapture(VIDEO_PATH)
n = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
fps = cap.get(cv2.CAP_PROP_FPS)
step = 10
sample_ids = list(range(0, n, step))
if sample_ids[-1] != n - 1:
    sample_ids.append(n - 1)
# Generate all downstream outputs only for sample_ids
```

Self-check list

- sample_ids strictly increasing, all < total frame count.
- Output coverage max index matches sample_ids[-1] (or matches your documented sampling policy).
- JSON keys are plain start->end, no extra text.
- Any per-frame artifact store (e.g., NPZ) contains exactly the sampled frames and no extras.